The Annals of Applied Statistics
2021, Vol. 15, No. 3, 1086–1100
https://doi.org/10.1214/21-AOAS1481
© Institute of Mathematical Statistics, 2021

A DEEP LEARNING SEMIPARAMETRIC REGRESSION FOR ADJUSTING COMPLEX CONFOUNDING STRUCTURES

BY XINLEI MI[1], PATRICK TIGHE[2], FEI ZOU[3,*] AND BAIMING ZOU[3,†]

[1]*Department of Preventive Medicine Biostatistics, Northwestern University, xinlei.mi@northwestern.edu*
[2]*Department of Anesthesiology, University of Florida, ptighe@anest.ufl.edu*
[3]*Department of Biostatistics, University of North Carolina at Chapel Hill, *feizou@email.unc.edu; †bzou@email.unc.edu*

Deep Treatment Learning (deepTL), a robust yet efficient deep learning-based semiparametric regression approach, is proposed to adjust the complex confounding structures in comparative effectiveness analysis of observational data, for example, electronic health record (EHR) data in which complex confounding structures are often embedded. Specifically, we develop a deep learning neural network with a score-based ensembling scheme for flexible function approximation. An improved semiparametric procedure is further developed to enhance the performance of the proposed method under finite sample settings. Comprehensive numerical studies have demonstrated the superior performance of the proposed methods, as compared with existing methods, with a remarkably reduced bias and mean squared error in parameter estimates. The proposed research is motivated by a postsurgery pain study, which is also used to illustrate the practical application of deepTL. Finally, an R package, "deepTL," is developed to implement the proposed method.

1. Introduction. The amount of electronic health record (EHR) data has expanded rapidly (Murdoch and Detsky (2013), Shah and Tenenbaum (2012), Psaty and Larson (2013)) and, inevitably, is used in various data-driven analyses in health care (Chen et al. (2013)). EHR data typically contain a large number of samples and often reflect daily clinical practice to offer valuable information on intervention efficacy under practical settings. Though comparative effectiveness analysis could be easily performed with randomized controlled trials (RCTs) (Britton et al. (1997), MacLehose et al. (2000), Benson and Hartz (2000)), in practice, RCTs cannot always be conducted for a variety of reasons (McCulloch et al. (2002), Curry, Reeves and Stringer (2003)). EHR data, on the other hand, are often readily available with rich information and serve as cost-effective alternatives to RCTs (Miriovsky, Shulman and Abernethy (2012)). However, the dependence among the treatment assignment, response and baseline characteristics can result in complicated confounding issues which can lead to biased estimation of intervention efficacy and misleading conclusions if they are not handled properly. In this paper we aim to perform valid comparative effectiveness analysis for EHR data with complex confounding structures.

In comparative effectiveness analysis a commonly used method to adjust for confounding factors is propensity score (PS) based methods (Rosenbaum and Rubin (1983)), including matching, covariate adjustment, stratification and inverse probability weighting (IPW) by PS. The PS methods use propensity scores to mimic RCTs such that samples with similar propensity scores have similar baseline features and thus are frequently used to analyze EHR data (Toh, García Rodríguez and Hernán (2011), Kazley and Ozcan (2008)). Under the strongly ignorable treatment assignment assumption, as shown by Rosenbaum and Rubin (1983), an unbiased estimate of the true treatment effect can be obtained by any of the PS-based methods. However, all PS-based methods heavily depend on the accuracy of the PS estimates, especially for PS-IPW and PS covariate adjustment (Austin (2011)).

Received June 2019; revised March 2020.
Key words and phrases. Bootstrap aggregating, comparative effectiveness analysis, complex confounding, deep neural network, propensity score, semiparametric regression.

For EHR data, confounding variables can impact the outcome and treatment allocation process in different ways with unknown functional formats, which makes PS modeling challenging. The motivating example in this paper is a postsurgery pain EHR data set (Tighe et al. (2016)). One of the study objectives is to compare the effectiveness of two anesthetic procedures, nerve block vs. general anesthesia, for relieving postsurgery pain intensity. Traditional methods, including a simple ANOVA analysis, a multivariate linear regression and a PS covariate adjustment method with PS estimated by a logistic regression, all lead to a nonsignificant difference between the two anesthesia groups (Table 3). However, previous findings in closely related postoperative pain studies under RCT designs suggest that the two groups are significantly different (Tverskoy et al. (1990), Shir, Raja and Frank (1994)). This raises awareness of the possibility that traditional parametric methods may fail to adequately detect complicated structures describing the connections among the pain intensity, the anesthesia and the covariates.

The performance of traditional parametric statistical methods heavily depends on their assumptions, such as the linearity assumption in the least square regression or logistic regression. To deal with the potential nonlinearity and other complexity in EHR data, nonparametric methods can be applied, such as kernel-based methods, including Nadaraya–Watson kernel estimators, Gaussian process models (Williams and Barber (1998)) and kernel-based support vector machines (SVMs). However, these local kernel-based machines are sensitive to the curse of dimensionality (Bengio, Delalleau and Roux (2006)). Though SVMs suffer less from the increase of dimensionality, due to sufficient regularization and postprocessing techniques for discrete outcomes (Platt et al. (1999)), it is argued that they may not be reliable for binary or multinomial outcomes (Tipping (2001)). In the recent literature the predictive modeling techniques for comparative effectiveness analysis have expanded and now include Lasso, gradient boosting machine, random forest and neural networks (Chernozhukov et al. (2016, 2018), Nie and Wager (2017)).

Apart from the PS framework, another commonly used strategy in comparative effectiveness analysis is to employ a semiparametric framework, as given below:

$$Y = \beta Z + \gamma(\boldsymbol{X}) + \epsilon, \tag{1.1}$$

where $(Y, Z, \boldsymbol{X})$ is the vector of the outcome, binary treatment assignment status and observed covariates; β is the treatment effect, γ is an unknown continuous function of $\boldsymbol{X}$ and $\epsilon \sim \mathrm{N}(0, \sigma^2)$. This semiparametric model has been widely investigated in the statistical literature (Engle et al. (1986), Robinson (1988), Stock (1991)). Robinson (1988) proposed an innovative strategy for obtaining an estimate of β with an optimal root-N-convergence rate. Instead of modeling equation (1.1) directly, Robinson (1988) proposed the following semiparametric model:

$$Y - \mathrm{E}(Y|\boldsymbol{X}) = \beta\big(Z - \mathrm{E}(Z|\boldsymbol{X})\big) + \epsilon. \tag{1.2}$$

With $\mathrm{E}(Y|\boldsymbol{X})$ and $\mathrm{E}(Z|\boldsymbol{X})$ preestimated from the Nadaraya–Watson kernel machine approach, a root-N-consistent estimate for β can be obtained via a simple linear regression model, based on Model (1.2) (Robinson (1988)).

Despite its root-N-consistency, Robinson's estimator has several limitations when applied to real-world EHR data. The performance of this semiparametric modeling strategy heavily depends on the accuracy of the estimation of $\mathrm{E}(Y|\boldsymbol{X})$ and $\mathrm{E}(Z|\boldsymbol{X})$. One obvious drawback is that the prediction accuracy of the Nadaraya–Watson kernel approach drops dramatically with the increase of the number of covariates, due to the curse of dimensionality (Bengio, Delalleau and Roux (2006)). In this paper, we employ the deep neural network (DNN), a fully-connected and feedforward neural network with multiple hidden layers, as a function approximator in the proposed framework of Robinson (1988) for comparative effectiveness

analysis. The neural network, a universally consistent function approximator, can approximate continuous functions on compact sets under certain assumptions (Cybenko (1989), Faragó and Lugosi (1993)). The strong universal consistency of the neural network offers a great potential to model complicated data, compared to traditional methods, such as logistic regression (Tu (1996)). To improve the accuracy and address the potential overfitting issue of the DNN, we further develop a score-based ensembling scheme via bootstrap aggregating (bagging) (Breiman (1996)).

Moreover, in modeling $\mathrm{E}(Y|\boldsymbol{X}) \equiv \xi(\boldsymbol{X})$, the residual $Y - \mathrm{E}(Y|\boldsymbol{X}) = \beta(Z - \mathrm{E}(Z|\boldsymbol{X})) + \epsilon$ no longer follows a unimodal distribution, especially when β deviates far from 0, which can potentially lead to inefficient estimation of $\xi(\boldsymbol{X})$. To minimize the impact from these limitations and offer an accurate estimate of the treatment effect, we propose a revised semiparametric procedure.

The rest of the paper is arranged as follows. A brief introduction of DNN implementation and an improved DNN ensemble model will first be presented in Section 2, followed by a detailed illustration of the modified semiparametric framework, deep treatment learning (deepTL). Extensive simulation studies and an application of the proposed method to a post-surgery pain study are presented in Section 3. Discussions and remarks conclude the paper in Section 4.

2. Deep treatment learning.

2.1. *Deep neural network.* For a general review of DNNs, we refer the readers to LeCun, Bengio and Hinton (2015). In this paper, we only introduce minimum but necessary concepts to facilitate the description of the proposed method.

We first introduce the general form of an L-hidden-layer feedforward DNN. The model contains L hidden layers of nodes that transform the initial input covariates $\boldsymbol{X}$ to the estimation of the output R, which can be a continuous or a binary response. For example, in our semiparametric regression framework, to model $\mathrm{E}(Z|\boldsymbol{X})$, $R = Z$ is a binary treatment assignment while modeling $\mathrm{E}(Y|\boldsymbol{X})$, $R = Y$ is a continuous response. For each hidden layer $l \in \{1, \dots L\}$, the model takes the input, denoted by $\boldsymbol{h}^{(l-1)}$, from the previous layer and the outputs $\boldsymbol{h}^{(l)}$ through function $\boldsymbol{g}_l(\boldsymbol{X})$. For any l, denote n_l as the number of elements in $\boldsymbol{h}^{(l)}$, then

$$\boldsymbol{h}^{(l)} = \boldsymbol{g}_l\big(\boldsymbol{b}^{(l)} + \boldsymbol{M}^{(l)}\boldsymbol{h}^{(l-1)}\big),$$

where $\boldsymbol{b}^{(l)}$ is the bias vector with length n_l and $\boldsymbol{M}^{(l)}$ is an $n_l \times n_{l-1}$ weight matrix. The function $\boldsymbol{g}_l$ is regarded as applying an activation function g_l elementwise to the n_l dimensional vector $\boldsymbol{b}^{(l)} + \boldsymbol{M}^{(l)}\boldsymbol{h}^{(l-1)}$. The activation function g_l is a nonlinear function that transforms the output values of neurons in the previous layer into the input values of the next layer. Often a common function g is applied to all g_l's ($l = 1, \dots, L$), for example, a rectified linear unit (ReLU) function (Hornik, Stinchcombe and White (1989)). For the first layer, that is, $l = 1$, $\boldsymbol{h}^{(0)}$ is simply the original p dimensional feature $\boldsymbol{X}$, and $\boldsymbol{g}_1$ takes a p dimensional input and produces an n_1 dimensional output. Finally, the Lth hidden layer $\boldsymbol{h}^{(L)}$ is tied to the output R through

$$R \sim g_{\mathrm{out}}\big(\boldsymbol{b}^{(L+1)} + \boldsymbol{M}^{(L+1)}\boldsymbol{h}^{(L)}\big),$$

where g_{out} gives a scalar output.

The output function is selected depending on the outcome R. For a continuous R we use an identity output function $g_{\mathrm{out}}(t) = t$; while for a binary R, a sigmoid output function $g_{\mathrm{out}}(t) = 1/(1 + e^{-t})$ is used.

The final convolved output function is

$$f = g_{\text{out}} \circ \boldsymbol{g}_L \circ \cdots \circ \boldsymbol{g}_1,$$

then f takes $\boldsymbol{X}$ as input and contains $\boldsymbol{\theta} = \{\boldsymbol{b}^{(i)}, \boldsymbol{M}^{(i)}\}_{i=1}^{L+1}$ as a collection of parameters. Under this setup, $\boldsymbol{\theta}$ can be estimated by minimizing the following empirical risk function:

$$\underset{\boldsymbol{\theta}}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^{N} \ell\{f(\boldsymbol{X}_i; \boldsymbol{\theta}); r_i\} + \lambda \Omega(\boldsymbol{\theta}),$$

where $\ell(\cdot;\cdot)$ is the loss function, $\Omega(\boldsymbol{\theta})$ is a penalty function and λ is a hyperparameter that controls the degree of regularization. For a continuous R we set the loss function $\ell(f, r)$ to $\frac{1}{2}(r - f)^2$. For a binary R, $\ell(f, r)$ is set to the Bernoulli negative log-likelihood, $-\{r \log f + (1 - r)\log(1 - f)\}$. Furthermore, to shrink the model size, we put an l_1 regularizer on the weight matrices $\boldsymbol{M}^{(l)}$ $(l = 1, \dots, L + 1)$ such that $\Omega(\boldsymbol{\theta}) = \sum_{l=1}^{L+1} \sum_{i=1}^{n_l} \sum_{j=1}^{n_{l-1}} |m_{ij}^{(l)}|$ is used, where $m_{ij}^{(l)}$ is the ijth element in the weight matrix $\boldsymbol{M}^{(l)}$.

We optimize the risk function by using the minibatch stochastic gradient descent algorithm (Byrd et al. (2012)), together with an adaptive learning rate adjustment method, that is, adaptive moment estimation (Adam) (Kinga and Adam (2015)).

2.2. *Bootstrap aggregating*. The total number of parameters in $\boldsymbol{\theta}$ is $N_{\boldsymbol{\theta}} = \sum_{l=1}^{L+1} n_l(n_{l-1} + 1)$, which is usually greater than the sample size N, leading to overparameterization and unstable prediction. The accuracy of the prediction from a single DNN model, therefore, is expected to be unreliable when the sample size is finite. An ensemble of neural networks has been shown to outperform a single neural network (Hansen and Salamon (1990)) in such scenarios. Thus, we apply bagging, that is, bootstrap aggregating (Breiman (1996)), to increase the robustness and accuracy of DNNs.

Specifically, we randomly sample the training set with replacement K times (i.e., bootstrap samples). Each time we fit a DNN model using a bootstrap sample and keep all the unused samples, namely, the out-of-bag (OOB) samples, as the validation set. Let $\widehat{f}_1(\cdot), \dots, \widehat{f}_K(\cdot)$ be the fitted models out of the K bootstrap samples. To aggregate the predictions from the K fitted models, a natural choice of the aggregated bagging prediction for a new observation with input $\boldsymbol{x}$ is $\widehat{f}(\boldsymbol{x}) = \sum_{k=1}^{K} \widehat{f}_k(\boldsymbol{x})/K$.

However, due to the randomness of the initial parameters, some DNNs may not converge to a stable solution and thus perform poorly. In neural network ensembles it is argued that "many could be better than all," meaning that using a subset of bagged DNNs that are well fit to the data could be better than using all of them (Zhou, Wu and Tang (2002), Mi, Zou and Zhu (2019)). Therefore, in our proposed method we remove certain DNNs, according to criteria defined below, which is consequently beneficial to the final ensemble model. For the kth bootstrap sample we define a performance score v_k as follows:

$$v_k = \begin{cases} \dfrac{1}{|D_{O_k}|} \displaystyle\sum_{i \in \mathcal{D}_{O_k}} \{(r_i - \overline{r}_{O_k})^2 - (r_i - \widehat{r}_{ik})^2\} & \text{for continuous } R, \\ \dfrac{1}{|D_{O_k}|} \displaystyle\sum_{i \in \mathcal{D}_{O_k}} \left\{ r_i \log\left(\frac{\widehat{r}_{ik}}{\overline{r}_{O_k}}\right) + (1 - r_i)\log\left(\frac{1 - \widehat{r}_{ik}}{1 - \overline{r}_{O_k}}\right) \right\} & \text{for binary } R, \end{cases}$$

where $\mathcal{D}_{O_k}$ is the set of OOB samples with $|D_{O_k}|$ being the associated sample size, $\overline{r}_{O_k} = \sum_{i \in \mathcal{D}_{O_k}} r_i / |D_{O_k}|$ and $\widehat{r}_{ik} = \widehat{f}_k(\boldsymbol{x}_i)$ $(i \in D_{O_k})$. For the regression DNN, v_k is the mean squared error loss, while, for the classification DNN, v_k can be interpreted as the negated binomial deviance.

To determine the optimal subset of DNNs retained in the ensemble, we first rank DNNs based on their performance scores, that is, $v_{(1)} \geq \cdots \geq v_{(K)}$. The prediction for $\boldsymbol{x}_i$ $(i = 1, \ldots, N)$ by aggregating the top q DNNs is then,

$$\widehat{f}^{(q)}(\boldsymbol{x}_i) = \frac{1}{q} \sum_{k \leq q} \widehat{f}_{(k)}(\boldsymbol{x}_i) \quad (q = 1, \ldots, K),$$

where $\widehat{f}_{(k)}(\cdot)$ is the fitted DNN corresponding to the performance score $v_{(k)}$. The optimal number of DNNs utilized by the ensemble, q_{opt}, is determined by minimizing the training loss, that is,

$$q_{\text{opt}} = \operatorname*{argmin}_{1 \leq q \leq K} \sum_{i=1}^{N} \ell\{\widehat{f}^{(q)}(\boldsymbol{x}_i), z_i\},$$

based on which we obtain the revised bagging prediction $\widehat{f}^{(q_{\text{opt}})}(\boldsymbol{x})$ for a new observation with input $\boldsymbol{x}$.

2.3. *Revised semiparametric regression.* Following (1.2), we let $\xi(\boldsymbol{X}) \equiv \mathrm{E}(Y|\boldsymbol{X})$ and $e(\boldsymbol{X}) \equiv \Pr(Z = 1|\boldsymbol{X}) = \mathrm{E}(Z|\boldsymbol{X})$. As Z in this paper is a binary treatment assignment status, $e(\boldsymbol{X})$ is the conditional probability for a subject being assigned to the treated group or the PS in Rosenbaum and Rubin (1983). Let $(Y_i, Z_i, \boldsymbol{X}_i)$ be the observed data of the ith sample $(i = 1, \ldots, N)$, and $\widehat{\xi}(\boldsymbol{X})$ and $\widehat{e}(\boldsymbol{X})$ be the corresponding approximation functions of $\xi(\boldsymbol{X})$ and $e(\boldsymbol{X})$, respectively, estimated from the observed data. Then, we can rewrite (1.2) as

$$Y - \widehat{\xi}(\boldsymbol{X}) = \beta\big(Z - \widehat{e}(\boldsymbol{X})\big) + \epsilon.$$

The estimate of β and its associated variance estimation can be obtained from the above simple linear regression model without an intercept as follows (Robinson (1988)):

$$\widehat{\beta} = \frac{\sum_{i=1}^{N}\{Y_i - \widehat{\xi}(\boldsymbol{X}_i)\}\{Z_i - \widehat{e}(\boldsymbol{X}_i)\}}{\sum_{i=1}^{N}\{Z_i - \widehat{e}(\boldsymbol{X}_i)\}^2}, \tag{2.1}$$

$$\widehat{V}(\widehat{\beta}) = \frac{\widehat{\sigma}^2}{\sum_{i=1}^{N}\{Z_i - \widehat{e}(\boldsymbol{X}_i)\}^2}, \tag{2.2}$$

where $\widehat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}[Y_i - \widehat{\xi}(\boldsymbol{X}_i) - \widehat{\beta}\{Z_i - \widehat{e}(\boldsymbol{X}_i)\}]^2$. Estimate $\widehat{\beta}$ is shown to be root-N-consistent under mild conditions, that is, $\sqrt{N}(\widehat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2\Phi^{-1})$ and $\widehat{\sigma}^2(\frac{1}{N}\sum_{i=1}^{N}\{Z_i - \widehat{e}(\boldsymbol{X}_i)\}^2)^{-1} \xrightarrow{p} \sigma^2\Phi^{-1}$, where $\Phi = \mathrm{E}\{Z - e(\boldsymbol{X})\}^2$ (Robinson (1988)).

With an infinite number of observations from $(Y, Z, \boldsymbol{X})$, $\widehat{\xi}(\boldsymbol{X})$ and $\widehat{e}(\boldsymbol{X})$ can consistently approximate $\xi(\boldsymbol{X})$ and $e(\boldsymbol{X})$, due to the universal consistency of DNNs (Faragó and Lugosi (1993), Hornik, Stinchcombe and White (1989), Sonoda and Murata (2017)). However, under finite-sample situations the model errors $\xi(\boldsymbol{X}) - \widehat{\xi}(\boldsymbol{X})$ and $e(\boldsymbol{X}) - \widehat{e}(\boldsymbol{X})$ exist. Note, the residual $Y - \xi(\boldsymbol{X}) = \beta(Z - e(\boldsymbol{X})) + \epsilon$, which is the sum of $\beta(Z - e(\boldsymbol{X}))$ and ϵ. When β is small, the sum is dominated by ϵ, which is Gaussian, and minimizing the mean squared difference of $\xi(\boldsymbol{X})$ from Y is equivalent to maximizing the log-likelihood of Gaussian random variables. This is expected to be efficient. However, when β is large, the distribution of the sum departs from a Gaussian distribution and is no longer unimodal, thus estimating $\xi(\boldsymbol{X})$ by minimizing the mean squared difference can become less efficient.

Accordingly, we propose to approach Model (1.1) by replacing (1.2) with the following modified model:

$$Y^* - \mathrm{E}\big(Y^*|\boldsymbol{X}, \beta_1\big) = (\beta - \beta_1)\big(Z - \mathrm{E}(Z|\boldsymbol{X})\big) + \epsilon, \tag{2.3}$$

Algorithm 1 deepTL: A Revised Semiparametric Regression

1: Approximate $e(\boldsymbol{X})$ by using a bagged-DNN approximator $\widehat{e}(\boldsymbol{X})$.
2: Estimate β_1 via the PS covariate adjustment method.
3: Approximate $\xi^*(\boldsymbol{X}, \beta_1)$ based on a bagged-DNN approximator $\widehat{\xi}^*(\boldsymbol{X})$.
4: Estimate β, that is, $\widehat{\beta}^*$, and $\mathrm{Var}(\widehat{\beta}^*)$ following (2.4) and (2.5).

where $Y^* = Y - \beta_1 Z$ for a given constant β_1. Instead of approximating $\xi(\boldsymbol{X})$, we approximate $\xi^*(\boldsymbol{X}, \beta_1) = \mathrm{E}(Y^*|\boldsymbol{X}, \beta_1)$ by our proposed score-based bagged DNNs in Section 2.2. The modified estimate of β and its associated variance estimation then become

$$\widehat{\beta}^* = \frac{\sum_{i=1}^N \{Y_i - \beta_1 Z_i - \widehat{\xi}^*(\boldsymbol{X}_i, \beta_1)\}\{Z_i - \widehat{e}(\boldsymbol{X}_i)\}}{\sum_{i=1}^N \{Z_i - \widehat{e}(\boldsymbol{X}_i)\}^2} + \beta_1 \quad \text{and} \tag{2.4}$$

$$\widehat{V}(\widehat{\beta}^*) = \frac{\widehat{\sigma^2}^*}{\sum_{i=1}^N \{Z_i - \widehat{e}(\boldsymbol{X}_i)\}^2}, \tag{2.5}$$

where $\widehat{\sigma^2}^* = \frac{1}{N}\sum_{i=1}^N [Y_i - \beta_1 Z - \widehat{\xi}^*(\boldsymbol{X}_i, \beta_1) - (\widehat{\beta}^* - \beta_1)\{Z_i - \widehat{e}(\boldsymbol{X}_i)\}]^2$.

Intuitively, if β_1 is close to β or when $|\beta - \beta_1|$ is small, the function approximation by bagged DNNs is expected to be more efficient which, subsequently, leads to an improved estimate of β. A reasonable candidate for β_1 can be an estimate of β from any PS approach in Rosenbaum and Rubin (1983). For the simulated data and real-data analysis, we set β_1 to the estimate derived from the PS covariate adjustment, where the estimated PS, $\widehat{e}(\boldsymbol{X})$ from bagged DNNs, is used. The accuracy of the estimated PS, $\widehat{e}(\boldsymbol{X})$, does not depend on β or the PS covariate adjustment which only depends on $\widehat{e}(\boldsymbol{X})$ (Rosenbaum and Rubin (1983)).

In summary, deepDL first approximates $e(\boldsymbol{X})$ by our proposed DNNs, based on which the treatment effect estimate is obtained from the PS covariate adjustment, and β_1 is subsequently set to this estimate. Next, we approximate $\xi^*(\boldsymbol{X}, \beta_1)$, again by our proposed DNNs, and obtain $\widehat{\xi}^*(\boldsymbol{X}, \beta_1)$. With $\widehat{e}(\boldsymbol{X})$ and $\widehat{\xi}^*(\boldsymbol{X}, \beta_1)$ obtained, we finally get $\hat{\beta}$ and its associated variance through (2.4) and (2.5). The modified estimator has the advantages of both the PS based method and the original semiparametric regression framework. The final algorithm is summarized in Algorithm 1.

3. Numerical studies. To investigate the performance of our proposed methods, we: (i) conduct extensive numerical studies under different confounding structures, from simple to complex settings, and (ii) perform a real-data analysis on a postsurgery pain data.

3.1. *Simulation studies*. We conduct simulation studies in three scenarios with different confounding structures. Specifically, we adopt the following models to generate data with three confounding structures under the semiparametric framework: $\boldsymbol{X} \sim \mathrm{MVN}(\boldsymbol{0}, I_p)$, $Z|\boldsymbol{X} \sim \mathrm{Bin}(1, e(\boldsymbol{X}))$, $Y = \beta Z + \gamma(\boldsymbol{X}) + \epsilon$, $\epsilon \sim \mathrm{N}(0, \sigma^2)$ and

$$\log\left(\frac{e(\boldsymbol{X})}{1 - e(\boldsymbol{X})}\right) = \begin{cases} 1 - \sum_{j=1}^5 \alpha_{1,j} X_j & \text{Scenario I,} \\ 1 - \sum_{j=1}^5 \alpha_{1,j} X_j + 2\cos(X_6) & \text{Scenario II,} \\ 1 - \sum_{j=1}^5 \alpha_{1,j} X_j - X_6^2 & \text{Scenario III;} \end{cases}$$

$$\gamma(\boldsymbol{X}) = \begin{cases} -1 + \sum_{j=1}^{5} \alpha_{2,j} X_j & \text{Scenario I,} \\ -1 + \sum_{j=1}^{5} \alpha_{2,j} X_j - \cos(2X_6) + X_7^2 & \text{Scenario II,} \\ -1 + \sum_{j=1}^{5} \alpha_{2,j} X_j + |X_6| X_7^2 & \text{Scenario III.} \end{cases}$$

The total sample size is set to $N = 5000$, and the dimension of $\boldsymbol{X}$, $p = 20$ with the treatment effect $\beta = 1$ or 2, the noise size $\sigma^2 = 1$. $\boldsymbol{\alpha}_1$ and $\boldsymbol{\alpha}_2$ are both five-valued vectors with elements $\alpha_{k,j} \sim U(-1, 1)$ i.i.d for $k = 1, 2$, $j = 1, \dots, 5$, which are generated at the beginning of Monte Carlo simulations and kept unchanged in the subsequent replications. All results are based on 1000 replicates. Besides the input and the output layers, all DNNs have six hidden layers with 20, 18, 16, 14, 12 and 10 hidden nodes from the first to the last hidden layers, respectively.

Besides deepTL, we also add semiDNN, which uses the same bagged DNNs as deepTL but directly implements Robinson's original semiparametric procedure (equations (2.1) and (2.2)), or, essentially, deepTL with β_1 being set to 0. In addition, we include an oracle linear regression method with underlying structures of $\mathrm{E}(Y|Z, \boldsymbol{X})$ known, denoted as "LM-Oracle." We also add "LM-Naive," a multiple linear regression method with all linear terms of the observed covariates and the treatment assignment status. We further include a PS covariate adjustment method, with the PS estimated by a logistic regression, denoted as "PS-Naive." Finally, the two cross-fitting double machine learning estimators, introduced in Chernozhukov et al. (2016), are included with all functions estimated by random forests, denoted as "DML-PLM" and "DML-DR." DML-PLM follows Robinson (1988) and adopts the original semiparametric regression framework, while DML-DR employs a double robust framework (Robins, Rotnitzky and Zhao (1994)).

For DNNs an l_1 penalty with weight $\lambda = 1e - 4$ is applied. The mini-batch stochastic gradient descent has a batch size $N_B = 100$, and Adam is employed with a starting learning rate of 0.001. The maximum number of epochs in DNN optimization is set to 250. The bagging size $K = 100$. All random forest models consist of 2000 trees.

To evaluate the performance of all methods, we provide the mean treatment effect estimate ($\mathrm{Avg}(\widehat{\beta})$), Monte Carlo standard error ($\mathrm{SE}(\widehat{\beta})$), estimated standard error ($\widehat{\mathrm{SE}}(\widehat{\beta})$), squared root of mean squared error ($\sqrt{\mathrm{MSE}}$) and the 95% confidence interval (95% CI) coverage.

Simulation results are summarized in Table 1. In Scenario I, because all the components are linear, LM-Naive is, essentially, the same as LM-Oracle. In this scenario, LM-Naive, PS-Naive, deepTL and LM-Oracle all estimate β unbiasedly. However, PS-Naive shows a 95% CI coverage of 98%, which is due to the inflated variance estimate of $\widehat{\beta}$ (Zou et al. (2016)). In addition, we observe a small bias in DML-PLM, when true $\beta = 2$, and thus a slightly increased MSE and a lower CI coverage than the targeted 95% coverage level. DML-DR appears to have a larger bias and Monte Carlo standard error and the largest MSE among all the methods. Among the two bagged DNN approaches the coverage of 95% CIs, provided by deepTL, is more precise than that of semiDNN. As the true β increases, we observe a larger bias and a remarkable decrease in 95% CI coverage in semiDNN, while the performance of deepTL remains almost unchanged with unbiased estimate and the targeted 95% CI coverage.

In Scenario II, as complex structures are introduced in $e(\boldsymbol{X})$ and $\gamma(\boldsymbol{X})$, naive methods, including LM-Naive and PS-Naive, fail, as expected, due to violations of the model assumptions. DML-PLM and DML-DR both show a reduced bias and Monte Carlo standard error, compared to naive methods. However, the biases from these methods are still not minor, leading to poor 95% CI coverages; semiDNN outperforms these methods. As expected, deepTL

TABLE 1
Simulation Results for Scenario I, II and III

Method	β	Avg($\widehat{\beta}$)	SE($\widehat{\beta}$)	$\widehat{\text{SE}}(\widehat{\beta})$	$\sqrt{\text{MSE}}$	95% CI	β	Avg($\widehat{\beta}$)	SE($\widehat{\beta}$)	$\widehat{\text{SE}}(\widehat{\beta})$	$\sqrt{\text{MSE}}$	95% CI
						Scenario I						
LM-Naive	1	1.000	0.035	0.034	0.035	94.8	2	2.001	0.035	0.034	0.035	93.8
PS-Naive		1.000	0.035	0.041	0.035	98.0		2.001	0.035	0.041	0.035	98.0
DML-DR		1.029	0.036	0.037	0.046	89.3		2.029	0.038	0.044	0.048	94.0
DML-PLM		1.001	0.034	0.035	0.034	96.3		1.986	0.036	0.035	0.038	92.7
semiDNN		0.986	0.035	0.034	0.038	92.8		1.969	0.036	0.035	0.047	84.8
deepTL		1.001	0.035	0.034	0.035	95.1		2.002	0.035	0.034	0.035	94.2
LM-Oracle		1.000	0.035	0.034	0.035	94.8		2.001	0.035	0.034	0.035	93.8
						Scenario II						
LM-Naive	1	0.587	0.059	0.059	0.417	0.0	2	1.584	0.058	0.059	0.420	0.0
PS-Naive		0.585	0.059	0.062	0.419	0.0		1.582	0.058	0.062	0.422	0.0
DML-DR		0.958	0.037	0.038	0.056	81.0		1.958	0.040	0.045	0.058	86.7
DML-PLM		0.944	0.035	0.040	0.066	72.2		1.905	0.039	0.040	0.103	33.5
semiDNN		0.987	0.036	0.035	0.038	92.3		1.972	0.035	0.035	0.045	88.3
deepTL		1.001	0.036	0.034	0.036	93.8		2.000	0.035	0.034	0.035	95.1
LM-Oracle		1.000	0.034	0.033	0.034	94.4		1.999	0.032	0.033	0.032	95.2
						Scenario III						
LM-Naive	1	1.529	0.058	0.057	0.533	0.0	2	2.528	0.059	0.057	0.532	0.0
PS-Naive		1.531	0.059	0.060	0.535	0.0		2.530	0.059	0.061	0.534	0.0
DML-DR		1.229	0.046	0.043	0.234	0.2		2.230	0.047	0.051	0.235	0.2
DML-PLM		1.141	0.040	0.044	0.146	7.5		2.109	0.041	0.044	0.116	29.2
semiDNN		0.987	0.036	0.036	0.039	92.7		1.971	0.038	0.037	0.048	86.7
deepTL		0.998	0.036	0.036	0.036	94.6		1.997	0.038	0.036	0.038	93.8
LM-Oracle		0.999	0.031	0.032	0.031	94.9		1.999	0.032	0.032	0.032	94.9

outperforms semiDNN, especially when $\beta = 2$, with an ignorable bias and a smaller standard error that is only slightly larger than that of LM-Oracle.

In Scenario III, deepTL and semiDNN continue to achieve a better performance than other competing methods, further indicating the robustness and advantages of our proposed methods. deepTL continues to perform similarly as LM-Oracle with a significant improvement in bias, compared to semiDNN. As more complex structure is introduced, DML-PLM and DML-DR show more severe biases and worse 95% CI coverages, and LM-Naive and PS-Naive continue to show a remarkably large bias.

To further investigate the performance of the proposed methods, we extend the simulations with different N, p, σ and β values under the same setting of Scenario III. Each time we vary one of the parameters and keep the rest of the parameters fixed. We exclude the results for LM-Naive and PS-Naive, due to their poor performance. The results are displayed in Figure 1.

First, we observe that the bias from semiDNN increases with the increment of β. In contrast, deepTL consistently offers an unbiased treatment effect estimate, regardless of the true treatment effect value. Next, as we increase N from 1000 to 20,000, as expected, all methods result in reduced biases and MSE. deepTL shows a negligible bias as N increases to 2000 or above, while the other methods still show noticeable biases at $N = 2000$. Moreover, as p increases from 10 to 50, semiDNN, DML-PLM and DML-DR all show increased biases, while deepTL is not affected as severely as the others. Finally, deepTL and semiDNN outperform DML-PLM and DML-DR regardless of the value of σ. In summary, deepTL performs better than, or equally as well, as other methods across all simulation parameters, reflecting its advantages with the modification of the semiparametric framework procedure and the use of bagged DNN models.

Additionally, to investigate how the model size, including L and n_l $(l = 1, \dots, L)$, and the number of DNNs, K, affect the performance of deepTL, more simulations under the setting of Scenario III for $\beta = 1$ are conducted. The results are presented in Figure 2. In the top plots, bagged DNNs have fixed $K = 100$ but L ranges from 1 to 6, n_1 ranges from 10 to 100 as the x-axis, $n_l = n_1 - \frac{l-1}{L-1}(n_1 - 10)$ for $l = 2, \dots, L$, while in the bottom plots, bagged DNNs are fixed with $L = 6$, $(n_1, \dots, n_6) = (20, 18, 16, 14, 12, 10)$ but K varies from 2 to 500.

Clearly, the impacts of L, n_1 and K on deepTL are minor. In the worst case, where $L = 2$, $n_1 = 10$ and $K = 100$, the absolute bias is 0.015, smaller than those from other competing methods, while, for most cases, the absolute bias of deepTL is less than 0.005. For MSE, we have similar observations. deepTL with $n_1 \in [20, 50]$ and $L \geq 3$, in general, performs well. Another encouraging finding is that the performance of deepTLincreases with the increase of K. However, the improvement becomes less obvious when $K > 100$, suggesting that deepTL only requires a moderate number of bagged DNNs.

3.2. *A postsurgery pain study.* To illustrate the practical application of deepTL, we apply it to a postsurgery pain data set obtained from the University of Florida Integrated Data Repository, a large-scale EHR database. The study (Tighe et al. (2016)) included 3196 patients who underwent different surgeries related to the digestive system, the musculoskeletal system and the nervous system. One of the objectives of the study was to compare two anesthetic procedures, that is, nerve block ($Z = 1$) s. general anesthesia ($Z = 0$), on relieving the severest postsurgery pain intensity within 72 hours after the surgery. Among these patients, 2438 (76.3%) patients chose the nerve block procedure while the remaining patients opted for general anesthesia. The nerve block procedure interrupts signals traveling along a nerve and is often used for pain relief. Compared to the traditional anesthesia procedure, the nerve block has some advantages by allowing patients to remain awake, thereby avoiding some adverse reactions of general anesthesia, such as cognitive loss. However, it is clinically important to test if the nerve block procedure is as effective as the general anesthesia. The primary outcome of the study was that the severest postsurgery pain intensity was experienced by patients

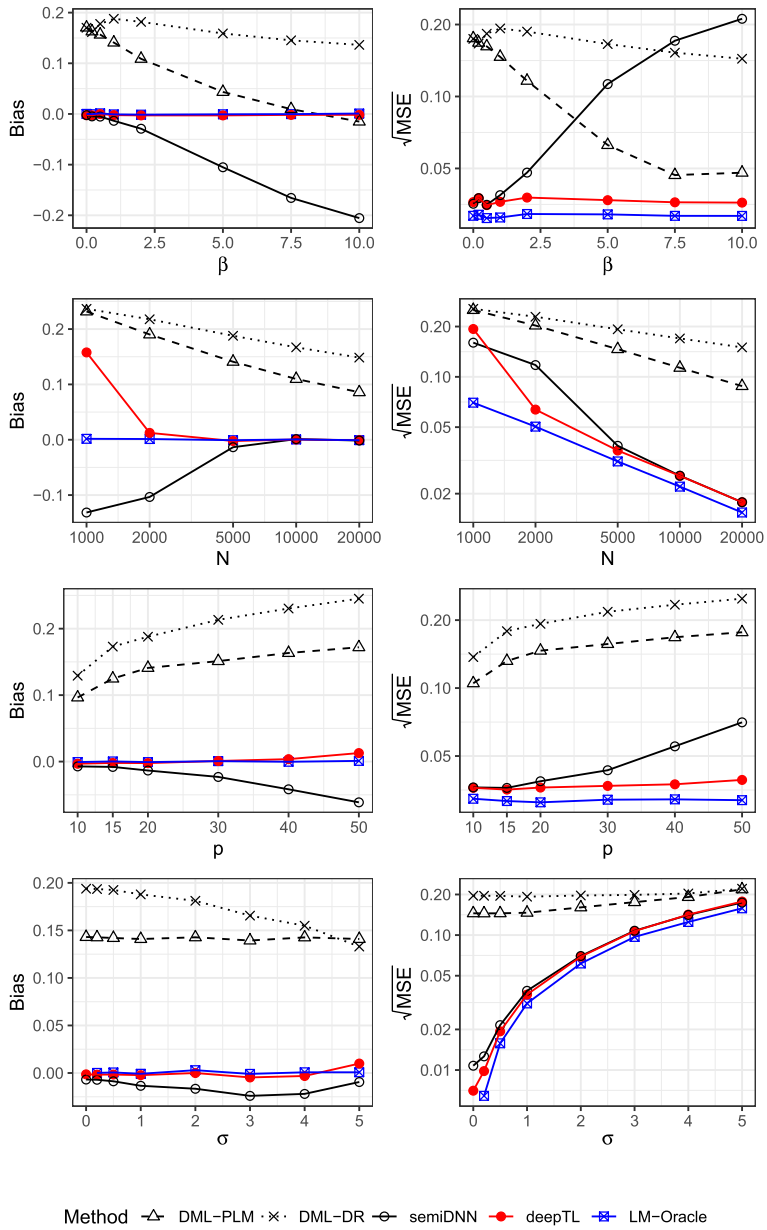

FIG. 1. *The bias and MSE plot for the simulation setting in Scenario III.*

within 72 hours after the surgery, which is quantified numerically and scaled between 0 and 10, where the higher pain scores mean more severe pain.

Covariates other than the treatment procedures (i.e., nerve block and general anesthesia) include patient age, gender, ethnicity, body mass index (BMI), surgical duration, marital status, opioid use, muscle relaxant use, nonsteroidal antiinflammatory drugs (NSAIDs) use, benzo use, selective serotonin reuptake inhibitors (SSRIs) use and current procedural terminology (CPT). Distributions of the baseline covariates by treatment group are presented in Table 2. We first apply an ANOVA method to the postoperative pain data and compare the effects of the two pain relief methods without any adjustment of the covariates. We also employ methods described in Section 3.1. The results are summarized in Table 3.

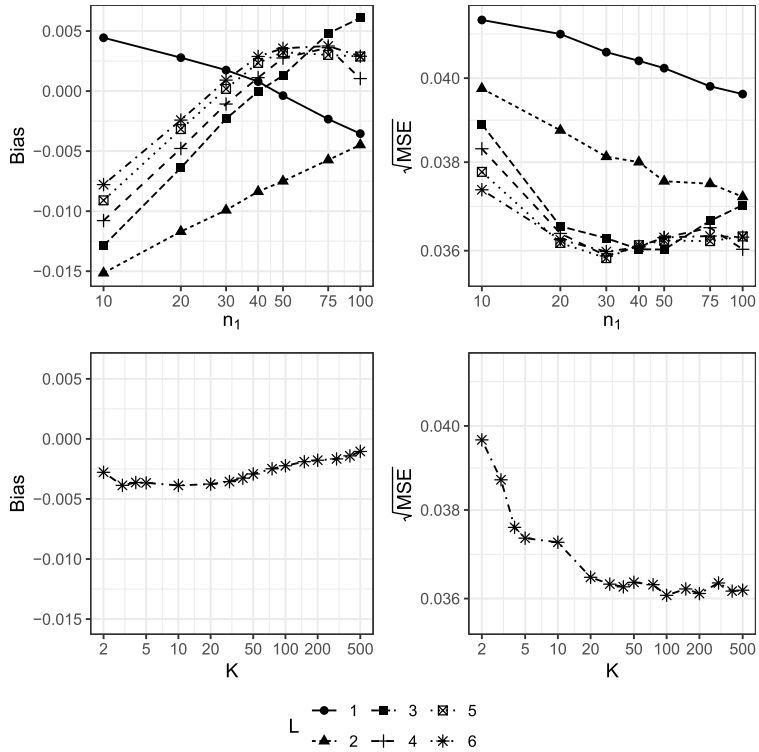

FIG. 2. *The bias and MSE plot for deepTL, using different L, n_1 and K, in bagged DNNs for the simulation setting in Scenario III when $\beta = 1$.*

The first row of Table 3 presents the crude treatment effect estimate, that is, -0.18, obtained from ANOVA. The nerve block is not significantly different from general anesthesia in this analysis ($p = 0.086$). A similar conclusion is obtained from LM-Naive and PS-Naive, even after the confounding covariates are adjusted, that is, there exists no significant difference between the two comparison procedures. Besides, DML-DR and DML-PLM show a less significant result. In contrast, the result from deepTL demonstrates that at the 0.05 significant level, the nerve block procedure is significantly more effective in relieving the severest post-surgery pain intensity than the general anesthesia procedure does ($p = 0.04$). This conclusion is also consistent with the results of earlier clinical studies (Tverskoy et al. (1990), Shir, Raja and Frank (1994)).

4. Discussion. In this paper, we proposed a powerful DNN-based semi-parametric framework, deepTL, to adjust the complex confounding structures in comparative effectiveness analysis. As a universally consistent approximator, DNN has a unique advantage over parametric supervised learning methods as well as other nonparametric machine learning approaches. In addition, bagging and the proposed ensembling scheme reduce the level of overfitting and increase the accuracy of DNN approximating functions, which consequently increases the treatment effect estimate accuracy (see Table S1 of Supplementary Material B (Mi et al. (2021)). Though the estimator from semi-DNN enjoys root-N-consistency, we have shown that the method can have an elevated bias under finite sample settings, when the underlying treatment effect β is not small, which motivates the development of deepTL. Extensive simulation studies demonstrate that deepTL consistently outperforms other existing competing methods for data with complex confounding structures, while, under simple settings, deepTL could still perform as well as other competing methods.

TABLE 2
Baseline Distribution of the Postsurgery Study Data

		Nerve Block	General
		2438 (76.3%)	758 (23.7%)
		Mean (SD)	Mean (SD)
Maximum pain intensity		8.0 (2.6)	8.2 (2.4)
Age (year)		58.1 (15.1)	55.4 (16.1)
BMI (kg/m^2)		29.6 (7.7)	29.2 (8.3)
Surgical Duration (hour)		3.9 (2.1)	3.1 (2.0)
		N (%)	N (%)
Gender	Male	1158 (47.5%)	360 (47.5%)
	Female	1280 (52.5%)	398 (52.5%)
Ethnicity	Non-Hispanic	2277 (93.4%)	716 (94.5%)
	Hispanic	49 (2.0%)	26 (3.4%)
	Unknown	112 (4.6%)	16 (2.1%)
	Married	1329 (54.5%)	381 (50.3%)
Marital	Single	602 (24.7%)	225 (29.7%)
	Other	507 (20.8%)	152 (20.1%)
Opioid	Yes	977 (40.1%)	296 (39.1%)
	No	1461 (59.9%)	462 (60.9%)
Muscle relaxant	Yes	204 (8.4%)	49 (6.5%)
	No	2234 (91.6%)	709 (93.5%)
NSAID	Yes	1418 (58.2%)	409 (54.0%)
	No	1020 (41.8%)	349 (46.0%)
Benzo	Yes	272 (11.2%)	89 (11.7%)
	No	2166 (88.8%)	669 (88.3%)
SSRI	Yes	609 (25.0%)	182 (24.0%)
	No	1829 (75.0%)	576 (76.0%)
	0–2	1952 (80.1%)	620 (81.8%)
CPT count	3–4	433 (17.8%)	127 (16.8%)
	$\geq$5	53 (2.2%)	11 (1.5%)

For observational data, in addition to confounding bias, overadjustment bias is another concern (Schisterman, Cole and Platt (2009)). To study overadjustment bias of deepTL, we carefully designed a new simulation setup, similar to DAG2 in Schisterman, Cole and Platt (2009). Under DAG2 we found that overadjustment bias is common to all comparing methods, while deepTL delivers the closest estimate to that from LM-Oracle (see Table S2 in

TABLE 3
Postoperative Pain Analysis

Method	$\widehat{\beta}$	$\widehat{\mathrm{SE}}(\widehat{\beta})$	p-value
ANOVA	−0.180	0.105	0.086
LM-Naive	−0.200	0.106	0.060
PS-Naive	−0.205	0.107	0.056
DML-DR	−0.079	0.218	0.718
DML-PLM	−0.138	0.105	0.188
semiDNN	−0.221	0.108	**0.040**
deepTL	−0.222	0.108	**0.040**

Supplement B). Furthermore, when there exist unmeasured confounding factors, the treatment effect estimate from deepTL as well as other existing methods can be biased in general. However, deepTL appears to control the bias to the minimal except for the oracle method (see Table S3 of Supplementary Material B (Mi et al. (2021) for simulation results on studying the robustness of deepTL on data with unmeasured confounding factors).

In deepTL, for computational efficiency we set β_1 to the estimate from the PS covariate adjustment method which works sufficiently well in improving the performance of deepTL over semi-DNN. Alternatively, one could set β_1 to the estimate from semi-DNN which is, however, more computationally demanding.

The proposed method is developed for continuous outcomes. However, it is not necessary for the DNN to be limited to this data type, though its performance for other data types, for example, binary outcomes, deserves further investigation. Furthermore, EHR data is often clustered and with repeated measurements from the same individual. How to extend the proposed method to nonindependent observational data is practically important as well but is beyond the scope of this paper.

Acknowledgments. We thank the Editor and anonymous reviewers for their insightful comments and constructive suggestions, which have greatly improved the paper.

SUPPLEMENTARY MATERIAL

Supplementary Material A: R package "deepTL" (DOI: 10.1214/21-AOAS1481SUPPA; .zip). We have developed an R package, Deep Treatment Learning "deepTL", to implement our proposed method.

Supplementary Material B: Additional simulation studies (DOI: 10.1214/21-AOAS1481SUPPB; .pdf). We provide additional supporting simulation studies to illustrate (i) necessity of the bagging procedure; (ii) the scenarios with overadjustment bias; (iii) the scenarios with unmeasured confounding.

REFERENCES

AUSTIN, P. C. (2011). An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivar. Behav. Res.* **46** 399–424. https://doi.org/10.1080/00273171.2011.568786

BENGIO, Y., DELALLEAU, O. and ROUX, N. L. (2006). The curse of highly variable functions for local kernel machines. In *Advances in Neural Information Processing Systems* 107–114.

BENSON, K. and HARTZ, A. J. (2000). A comparison of observational studies and randomized, controlled trials. *N. Engl. J. Med.* **342** 1878–1886.

BREIMAN, L. (1996). Bagging predictors. *Mach. Learn.* **24** 123–140.

BRITTON, A., MCKEE, M., BLACK, N., MCPHERSON, K., SANDERSON, C. and BAIN, C. (1997). Choosing between randomised and non-randomised studies: A systematic review. *Health Technology Assessment* (*Winchester, England*) **2** i–iv.

BYRD, R. H., CHIN, G. M., NOCEDAL, J. and WU, Y. (2012). Sample size selection in optimization methods for machine learning. *Math. Program.* **134** 127–155. MR2947555 https://doi.org/10.1007/s10107-012-0572-5

CHEN, Y., CARROLL, R. J., HINZ, E. R. M., SHAH, A., EYLER, A. E., DENNY, J. C. and XU, H. (2013). Applying active learning to high-throughput phenotyping algorithms for electronic health records data. *J. Am. Med. Inform. Assoc.* **20** e253–e259.

CHERNOZHUKOV, V., CHETVERIKOV, D., DEMIRER, M., DUFLO, E., HANSEN, C. and NEWEY, W. K. (2016). Double machine learning for treatment and causal parameters. Technical Report No. CWP49/16. CeMMAP working paper, Centre for Microdata Methods and Practice, London.

CHERNOZHUKOV, V., CHETVERIKOV, D., DEMIRER, M., DUFLO, E., HANSEN, C., NEWEY, W. and ROBINS, J. (2018). Double/debiased machine learning for treatment and structural parameters. *Econom. J.* **21** C1–C68. MR3769544 https://doi.org/10.1111/ectj.12097

CURRY, J. I., REEVES, B. and STRINGER, M. D. (2003). Randomized controlled trials in pediatric surgery: Could we do better? *Journal of Pediatric Surgery* **38** 556–559.

CYBENKO, G. (1989). Approximation by superpositions of a sigmoidal function. *Math. Control Signals Systems* **2** 303–314. MR1015670 https://doi.org/10.1007/BF02551274

ENGLE, R. F., GRANGER, C. W., RICE, J. and WEISS, A. (1986). Semiparametric estimates of the relation between weather and electricity sales. *J. Amer. Statist. Assoc.* **81** 310–320.

FARAGÓ, A. and LUGOSI, G. (1993). Strong universal consistency of neural network classifiers. *IEEE Trans. Inf. Theory* **39** 1146–1151.

HANSEN, L. K. and SALAMON, P. (1990). Neural network ensembles. *IEEE Trans. Pattern Anal. Mach. Intell.* **12** 993–1001.

HORNIK, K., STINCHCOMBE, M. and WHITE, H. (1989). Multilayer feedforward networks are universal approximators. *Neural Netw.* **2** 359–366.

KAZLEY, A. S. and OZCAN, Y. A. (2008). Do hospitals with electronic medical records (EMRs) provide higher quality care?: An examination of three clinical conditions. *Med. Care Res. Rev.* **65** 496–513. https://doi.org/10.1177/1077558707313437

KINGA, D. and ADAM, J. B. (2015). A method for stochastic optimization. In *International Conference on Learning Representations* (*ICLR*) **5**.

LECUN, Y., BENGIO, Y. and HINTON, G. (2015). Deep learning. *Nature* **521** 436–444.

MACLEHOSE, R., REEVES, B., HARVEY, I., SHELDON, T., RUSSELL, I. and BLACK, A. (2000). A systematic review of comparisons of effect sizes derived from randomised and non-randomised studies. *Health Technology Assessment* (*Winchester, England*) **4** 1–154.

MCCULLOCH, P., TAYLOR, I., SASAKO, M., LOVETT, B. and GRIFFIN, D. (2002). Randomised trials in surgery: Problems and possible solutions. *BMJ, Br. Med. J.* **324** 1448–1451.

MI, X., TIGHE, P., ZOU, F. and ZOU, B. (2021). Supplement to "A deep learning semiparametric regression for adjusting complex confounding structures." https://doi.org/10.1214/21-AOAS1481SUPPA, https://doi.org/10.1214/21-AOAS1481SUPPB

MI, X., ZOU, F. and ZHU, R. (2019). Bagging and deep learning in optimal individualized treatment rules. *Biometrics* **75** 674–684. MR3999189 https://doi.org/10.1111/biom.12990

MIRIOVSKY, B. J., SHULMAN, L. N. and ABERNETHY, A. P. (2012). Importance of health information technology, electronic health records, and continuously aggregating data to comparative effectiveness research and learning health care. *J. Clin. Oncol.* **30** 4243–4248.

MURDOCH, T. B. and DETSKY, A. S. (2013). The inevitable application of big data to health care. *JAMA* **309** 1351–1352.

NIE, X. and WAGER, S. (2017). Learning objectives for treatment effect estimation. Preprint. Available at arXiv:1712.04912.

PLATT, J. et al. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers* **10** 61–74.

PSATY, B. M. and LARSON, E. B. (2013). Investments in infrastructure for diverse research resources and the health of the public. *JAMA* **309** 1895–1896.

ROBINS, J. M., ROTNITZKY, A. and ZHAO, L. P. (1994). Estimation of regression coefficients when some regressors are not always observed. *J. Amer. Statist. Assoc.* **89** 846–866. MR1294730

ROBINSON, P. M. (1988). Root-N-consistent semiparametric regression. *Econometrica* **56** 931–954. MR0951762 https://doi.org/10.2307/1912705

ROSENBAUM, P. R. and RUBIN, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika* **70** 41–55. MR0742974 https://doi.org/10.1093/biomet/70.1.41

SCHISTERMAN, E. F., COLE, S. R. and PLATT, R. W. (2009). Overadjustment bias and unnecessary adjustment in epidemiologic studies. *Epidemiology* (*Cambridge, Mass.*) **20** 488–495.

SHAH, N. H. and TENENBAUM, J. D. (2012). Focus on translational bioinformatics: The coming age of data-driven medicine: Translational bioinformatics' next frontier. *J. Am. Med. Inform. Assoc.* **19** e2.

SHIR, Y., RAJA, S. N. and FRANK, S. M. (1994). The effect of epidural versus general anesthesia on postoperative pain and analgesic requirements in patients undergoing radical prostatectomy. *Anesthesiology* **80** 49–56.

SONODA, S. and MURATA, N. (2017). Neural network with unbounded activation functions is universal approximator. *Appl. Comput. Harmon. Anal.* **43** 233–268. MR3668038 https://doi.org/10.1016/j.acha.2015.12.005

STOCK, J. H. (1991). Nonparametric policy analysis: An application to estimating hazardous waste cleanup benefits. In *Nonparametric and Semiparametric Methods in Econometrics and Statistics* 77–98. Cambridge University Press, Cambridge.

TIGHE, P. J., KING, C. D., ZOU, B. and FILLINGIM, R. B. (2016). Time to onset of sustained postoperative pain relief (SuPPR): Evaluation of a new systems-level metric for acute pain management. *The Clinical Journal of Pain* **32** 371–379.

TIPPING, M. E. (2001). Sparse Bayesian learning and the relevance vector machine. *J. Mach. Learn. Res.* **1** 211–244. MR1875838 https://doi.org/10.1162/15324430152748236

TOH, S., GARCÍA RODRÍGUEZ, L. A. and HERNÁN, M. A. (2011). Confounding adjustment via a semi-automated high-dimensional propensity score algorithm: An application to electronic medical records. *Pharmacoepidemiol. Drug Saf.* **20** 849–857.

TU, J. V. (1996). Advantages and disadvantages of using artificial neural networks versus logistic regression for predicting medical outcomes. *J. Clin. Epidemiol.* **49** 1225–1231.

TVERSKOY, M., COZACOV, C., AYACHE, M., BRADLEY, J. E. and KISSIN, I. (1990). Postoperative pain after inguinal herniorrhaphy with different types of anesthesia. *Anesthesia and Analgesia* **70** 29–35.

WILLIAMS, C. K. and BARBER, D. (1998). Bayesian classification with Gaussian processes. *IEEE Trans. Pattern Anal. Mach. Intell.* **20** 1342–1351.

ZHOU, Z.-H., WU, J. and TANG, W. (2002). Ensembling neural networks: Many could be better than all. *Artificial Intelligence* **137** 239–263. MR1906477 https://doi.org/10.1016/S0004-3702(02)00190-X

ZOU, B., ZOU, F., SHUSTER, J. J., TIGHE, P. J., KOCH, G. G. and ZHOU, H. (2016). On variance estimate for covariate adjustment by propensity score analysis. *Stat. Med.* **35** 3537–3548. MR3537221 https://doi.org/10.1002/sim.6943